

Query-Aware Visual Token Pruning as a Diagnostic Probe of Multimodal Attention

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Abstract

Multimodal large language models (MLLMs) process images as long sequences of visual tokens, making inference cost sensitive to token count. Prior work has shown that visual token pruning can improve efficiency, but it remains unclear when attention-derived pruning scores correspond to semantically necessary visual evidence rather than merely serving as a useful heuristic. We study this question by using token pruning as a diagnostic probe of multimodal attention. We compare attention-based visual token retention against no compression, random token dropping, uniform spatial selection, and inverse-attention pruning across retention rates and attention layers. On a controlled GQA-style evaluation, middle-layer attention yields the strongest efficiency-accuracy frontier: at aggressive retention levels it preserves substantially more accuracy than early-layer attention and strongly outperforms inverse pruning. Category-level analysis shows that attention-based retention is most reliable for object, attribute, and global-scene queries, but weaker for spatial reasoning and underrepresented counting cases. These findings support a conditional version of the hypothesis: attention-derived relevance can identify useful visual tokens, but its reliability depends on layer depth, question type, and the distribution of visual evidence.

1. Introduction

Multimodal large language models (MLLMs) such as LLaVA and Qwen have become a dominant architecture for visual question answering and image-grounded dialogue. A common design pattern is to encode an image into a sequence of visual tokens and pass those tokens to a large language model, either through cross-attention modules, learned query tokens, or direct projection into the language model embedding space. This design is powerful but expensive: as image resolution or tiling increases, the number of visual tokens grows, and the language model must process more context during prefilling and generation. For deployment, this creates an important efficiency problem. For research, it raises a deeper mechanistic question: are all visual tokens equally necessary for a particular query, or does the model’s own attention reveal which image regions are relevant?

Visual token pruning has emerged as a practical answer to the efficiency problem. Recent methods reduce visual tokens using attention scores, instruction guidance, token similarity, or diversity constraints. However, showing that a pruning rule improves the accuracy-compute trade-off is not the same

as showing that the rule captures a meaningful deep learning mechanism. If an attention-based method beats a full-token baseline only marginally, or if simple random or spatial baselines perform similarly, the result may reflect redundancy in visual tokens rather than faithful query-conditioned relevance. Conversely, if inverse-attention pruning consistently fails while attention pruning succeeds, this provides stronger evidence that the ranking contains useful information about visual evidence.

This paper studies visual token pruning as a diagnostic probe rather than merely as an acceleration technique. We ask: when do attention-derived, query-conditioned relevance scores identify semantically useful visual tokens better than query-agnostic baselines? Our hypothesis is that attention-based retention should work best when the evidence required by the query is localized or strongly aligned with salient visual regions, but should be less reliable when the answer depends on distributed, relational, or spatial evidence. We test this hypothesis using controlled pruning experiments across retention rates, attention layers, and question categories.

2. Related work

2.1. Multimodal language models and visual tokens

Modern MLLMs differ in how visual representations are connected to language models. Flamingo inserts gated cross-attention layers into a frozen language model, allowing text tokens to condition on visual inputs [2]. BLIP-2 introduces a Querying Transformer (Q-Former) to bridge a frozen vision encoder and a frozen language model [11]. LLaVA demonstrates that a simpler architecture, in which CLIP visual features are projected into the language token space and instruction-tuned, can produce strong multimodal dialogue ability [13, 14]. Qwen scales the vision backbone and uses dynamic-resolution processing, highlighting the importance of token allocation for high-resolution understanding [6, 7]. These architectures make visual tokens a central computational bottleneck.

2.2. Visual token reduction

The broader idea of token reduction predates MLLMs. In vision transformers, DynamicViT prunes tokens dynamically based on input-dependent sparsification [16], while Token Merging (ToMe) merges similar tokens to reduce computation without retraining [4]. These methods show that visual representations contain redundancy, but they are generally

query-agnostic: the image is compressed without considering the downstream language question.

MLLM-specific pruning methods introduce language or attention guidance. FastV observes that visual tokens become less necessary after early layers and prunes them based on attention patterns [5]. IVTP performs instruction-guided visual token pruning in two stages, using vision-side attention rollout and language-side instruction relevance [8]. CATP ranks visual tokens using cross-attention in multimodal inference [12], while LVPruning uses language-guided token importance in existing MLLMs without modifying base parameters [17]. SparseVLM and FitPrune show that training-free pruning can also be competitive by exploiting attention statistics or attention-distribution preservation [21, 20]. More recent diversity-aware methods argue that importance alone may be insufficient and that retained tokens should also preserve coverage of the visual scene [3, 22].

2.3. Attention as explanation

The use of attention as a relevance signal is controversial. Jain and Wallace argue that attention weights should not automatically be treated as explanations because alternative attention distributions may yield similar predictions [10]. Abnar and Zuidema show that raw attention in transformers is difficult to interpret because information is repeatedly mixed across layers, motivating attention rollout and attention flow as more faithful approximations to input relevance [1]. Wiegrefe and Pinter offer a more nuanced view: attention can be explanatory under carefully specified diagnostic assumptions, but it must be evaluated rather than assumed [19]. This debate directly motivates our work. If attention is used for visual token pruning, then attention-based token scores implicitly claim something about visual importance. Our experiments test that claim by comparing against query-agnostic and inverse-attention baselines.

2.4. Positioning

We do not claim novelty in visual token pruning itself. Instead, we use pruning as a controlled perturbation to study when attention-derived token importance aligns with useful visual evidence. This positioning matches recent concerns that token pruning papers should compare against simple baselines and analyze when pruning fails, rather than reporting efficiency alone [18]. Our study is therefore closest in spirit to a mechanism-oriented empirical analysis: the goal is to understand how attention, visual redundancy, and query structure interact.

3. Method

3.1. Problem formulation

Let an MLLM receive an image I and language query q . The image encoder maps I to visual tokens $V = \{v_1, \dots, v_N\}$, which are projected or otherwise supplied to the language model. A pruning method selects a subset $S \subseteq V$ of size $|rN|$, where $r \in (0, 1]$ is the retention rate. The model then answers using only the retained tokens. The central question

is whether the selection rule preserves task-relevant visual evidence.

We compare four selection rules. The first is no compression, which keeps all tokens and provides the reference accuracy. The second is random retention, which samples $|rN|$ visual tokens uniformly at random. The third is uniform spatial retention, which keeps tokens according to a fixed spatial pattern, approximating query-agnostic coverage. The fourth is attention-based retention, which ranks visual tokens by their attention-derived relevance to the query.

3.2. Attention-derived token scores

For LLaVA-style models, visual and text tokens are processed in a shared autoregressive transformer. Since there may not be an explicit cross-attention module, we use text-to-visual self-attention within the joint sequence as the relevance signal. For a chosen layer ℓ , attention head set H , text token set T , and visual token v_i , we define an importance score

$$s_i^{(\ell)} = \frac{1}{|H||T|} \sum_{h \in H} \sum_{t \in T} A_h^{(\ell)}(t, i), \quad (1)$$

where $A_h^{(\ell)}(t, i)$ is the attention from text position t to visual token position i . The two pipelines instantiate T differently. On LLaVA we restrict T to the final text-prompt position immediately preceding generation, following the FastV operationalisation [5]; this aligns the score with the model’s most recent query-conditioned state and avoids confounding the signal with attention paid to instruction-template tokens. On Qwen2.5-VL we instead average across all non-visual prompt positions, which provides a denser query-conditioned signal at the cost of giving some weight to instruction-template attention. The two aggregations are different operationalisations of the same principle, and reporting both models under the same selectors lets us check that the cross-method ordering is robust to this design choice. Tokens are sorted by $s_i^{(\ell)}$ and the top $|rN|$ are retained.

3.3. Inverse-attention sanity check

A key weakness of many pruning studies is that improved efficiency alone does not establish that the scoring rule is meaningful. We therefore include an inverse-attention baseline, which retains the tokens with the lowest attention-derived scores. If attention truly carries information about token usefulness, inverse pruning should perform substantially worse than top-attention pruning, especially at low retention rates. If inverse pruning performs similarly, then the attention ranking is not a reliable relevance signal.

3.4. Implementation details

Selection logic. The selection logic is shared across both models. For each image-query pair, the model first executes a forward pass at the chosen layer to obtain the attention tensor, from which the importance score in Equation 1 is computed over the visual token positions. The top or bottom $|rN|$ scores are retained depending on whether top-attention or inverse-attention selection is in use. Random retention

uses a fixed seed for reproducibility, and uniform retention takes evenly spaced indices across the visual sequence so that the selected tokens cover the image grid as uniformly as possible. All methods are evaluated on the same examples and with the same answer-matching routine.

Token reduction mechanisms. The two models in our study realise the actual token reduction through complementary mechanisms, a deliberate choice that strengthens the cross-architecture interpretation. On LLaVA we use *attention masking*: the visual token positions remain in the input sequence, and the entries of the attention mask corresponding to non-retained positions are set to zero before generation. The sequence length is therefore unchanged at runtime, so the reported compute proxy (Section 4.6) is analytical rather than measured, but the only quantity that varies across pruning conditions is which visual tokens contribute to attention. This isolates the scientific question—“does the ranking carry useful relevance information?”—from engineering details such as cache reshaping, position renumbering, or layer-conditioned drop schedules. On Qwen we instead apply *physical removal at the embedding level*: the visual encoder and projection merger run at full cost to produce the full visual embedding sequence, the selected subset is extracted from this sequence, and the language decoder generates from a rebuilt input-embedding sequence consisting of the text prefix, the kept visual embeddings, and the text suffix. The decoder sequence length is therefore reduced in proportion to the retention rate. Running the same selection rules under both an attention-masking implementation and a physical-removal implementation provides cross-implementation evidence that any observed method ordering is not an artefact of either choice alone.

Loading and precision. Both models are loaded in 4-bit precision with float16 compute. On LLaVA we additionally enable double quantisation and use the eager attention implementation, the latter being required to materialise the attention matrix needed for scoring. On Qwen the attention tensor is materialised at inference time through the model’s standard output-attention interface. Every pruning condition is evaluated on the same quantised model and the same example set within each pipeline, so the relative ordering of methods, which is the quantity of scientific interest, is preserved.

Aggregation simplicity. The attention aggregation rule is intentionally simple because the experiment is diagnostic. A more elaborate method could combine rollout, gradients, diversity, or learned gates, but doing so would make it harder to isolate what raw attention contributes. By averaging over heads at a single layer and using either the final query position or the full text-prompt span, the experiment asks a direct question: is the attention pattern already present in the model sufficient to rank visual tokens in a useful way?

3.5. Expected failure modes

Before running the experiments, we expect three types of failure. First, attention may be positionally biased, causing the method to retain tokens from common image regions even when the query refers elsewhere. Second, attention may be semantically incomplete: a token can receive high attention because it is salient, while a low-attention token still contains evidence needed for a comparison or negation. Third, attention may be layer-misaligned. Early layers may not yet encode query-conditioned relevance, while very late layers may encode answer-level shortcuts rather than visual grounding. These failure modes guide the ablations and discussion.

4. Experiments

4.1. Models and inference setup

Our experiments use two multimodal models with substantially different vision-token regimes, which provides a useful contrast for the cross-architecture analysis. The first is LLaVA-1.5-7B [14], which encodes images with CLIP-ViT-L/14 at 336×336 resolution and produces a fixed sequence of $N = 576$ visual tokens projected into a 32-layer Vicuna-7B decoder of hidden dimension 4096. The second is Qwen2.5-VL-3B-Instruct [6, 7], whose visual encoder operates on a larger raw-patch grid which a learned projection merger then downsamples to $N = 393$ visual embeddings before the language decoder; the Qwen decoder has 36 layers and hidden dimension 2048. Pruning is implemented via attention masking on LLaVA and via physical removal of non-retained visual embeddings on Qwen, as described in Section 3.4. On LLaVA we use a standard single-turn instruction template that places the image marker before the question and elicits a short-form answer; on Qwen we use the native chat template provided by its processor, which delimits the visual span with explicit vision-start and vision-end marker tokens. Decoding is greedy in both pipelines, with a 20-token generation cap on GQA and a 10-token cap on MMBench. Both models are evaluated under identical retention rates, selectors, layer choices, and stratified example sets, so reported cross-model differences reflect architecture and implementation rather than evaluation setup.

4.2. Datasets and evaluation

We evaluate primarily on the GQA testdev-balanced split [9], using a fixed 1,000-example stratified subset drawn under a controlled seed and balanced across GQA’s five structural question types: 350 query, 200 verify, 150 logical, 150 choose, and 150 compare examples. The same subset is used under every pruning condition and across both models, so that paired analysis across methods is possible on identical examples. We additionally include a 200-example random subset of MMBench-dev, English split [15], drawn under the same seed, as a cross-format sanity check to verify that the cross-method ordering observed on open-answer GQA also holds in the multiple-choice perception-and-reasoning setting. For each condition we store a prediction file containing

the example identifier, question, ground-truth answer, normalised prediction, structural type, and semantic category.

We report accuracy as the main task metric. For efficiency, we use the retained token percentage and an approximate per-example GFLOPs estimate. GFLOPs should be interpreted as a compute proxy rather than a hardware-specific latency measure.

4.3. Methods compared

We compare a full-token baseline, random retention, uniform spatial retention, attention-based retention from the language decoder, and inverse retention from the same source layer. The attention method was profiled across a four-layer candidate set spanning early, early-mid, mid, and late decoder depths, namely $\ell \in \{2, 8, 15, 22\}$, which is applied at the same indices on both models. The main paper focuses on layers 2 and 15 because these two settings most clearly separate shallow positional attention from query-conditioned mid-layer attention, and the full layer sweep is summarised in Section 7. The inverse probe is run only at the best-performing layer, $\ell = 15$. Retention rates include 90%, 70%, 50%, 25%, and 10% where available; the full-token baseline is evaluated at 100% retention.

4.4. Question category analysis

To test the hypothesis that pruning reliability depends on the structure of the query, we group GQA examples by semantic category at 25% retention. The five-class taxonomy used in Section 6—object recognition, attribute, counting, spatial reasoning, and global scene—is derived from a combination of GQA’s native semantic-type metadata and a small set of keyword cues over the question text. Counting examples are identified by the presence of standard quantifier phrases such as “how many” or “number of”. Spatial examples are taken from GQA’s relational semantic type and augmented with explicit relational vocabulary, including terms such as left, right, above, below, behind, and in front of. Global scene and attribute examples are taken directly from the corresponding GQA semantic types, and remaining examples, predominantly identification queries, are grouped under object recognition. Counting is discussed as a limitation rather than reported per method, because the stratified subset yielded too few counting examples on either model to support a reliable category-level estimate.

4.5. Answer matching and normalization

Open-ended VQA evaluation is sensitive to surface-form variation, and the two models exhibit different verbosity profiles: LLaVA tends to produce short answers consistent with the single-word-or-phrase instruction, while Qwen2.5-VL frequently returns full sentences even on yes/no questions, such as “Yes, there is a couch in the photo.” rather than a bare “yes.” We therefore use a model-aware matching rule. Predictions are first normalised by lowercasing, stripping whitespace, and removing punctuation. For LLaVA, accuracy is computed by exact match between the normalised prediction and the normalised gold answer. For Qwen, exact

match is used when the gold answer is multi-token, but for yes/no gold answers we instead check whether the gold token appears as a whole word in the normalised prediction; this avoids penalising surface-form verbosity that does not affect semantic correctness. For MMBench, which uses multiple-choice options A through D, we extract the first option letter that appears in the generated text and compare it against the gold letter. Records whose prediction is an explicit generation or out-of-memory error are excluded from the accuracy calculation, and we track the residual error rate per condition, reporting only conditions whose error rate is below a five-percent threshold. This matching rule is sufficient for the comparative purpose of the experiment because every pruning condition is evaluated on the same set of questions; it may still under-count semantically equivalent free-form answers such as “woman” versus “lady”, which we treat as a source of conservative noise rather than systematic bias.

4.6. Compute proxy

For both pipelines we report an analytical GFLOPs-per-example estimate derived from the retained-token count $k = \lfloor rN \rfloor$ at the language-decoder level. The estimate uses standard transformer cost expressions for the prefill stage, applied to a combined text-and-visual prompt of length $T_{\text{text}} + k$, together with an analogous decode-stage cost over a short generation horizon. The two models supply different architectural constants. On LLaVA we use hidden dimension 4096, feed-forward dimension 11008, 32 decoder layers, 576 visual tokens, a 60-token prompt budget, and a 20-token decode budget; on Qwen we use hidden dimension 2048, feed-forward dimension 11008, 36 decoder layers, and 393 visual tokens, with the same prompt and decode budgets. Two interpretive caveats apply. First, on LLaVA pruning is realised via attention masking (Section 3.4), so the language-decoder sequence length is unchanged at runtime; the reported GFLOPs therefore represent the cost a physical-removal implementation would incur at the same retention rate, not a measured wall-clock saving. Second, on Qwen physical removal does shorten the sequence seen by the decoder, so the analytical GFLOPs more closely approximate the true decoder-side compute saving; however, the visual encoder and merger run at full cost regardless of retention rate, and absolute speedup additionally depends on cache layout, batching, and kernel choice. The axis is therefore comparable across models in the sense that both pipelines apply the same retained-token-count formula to the same decoder cost expressions; the interpretation of what was saved differs.

5. Main Results

Table 1 compares the five strategies on GQA at representative retention ratios. Full results including MMBench accuracy are provided in Appendix A.

LLaVA-1.5-7B.

On LLaVA-1.5-7B, attention-based pruning at $l=15$ is the strongest method across all retention rates. At $r=0.25$, it

Table 1: GQA accuracy (%) across token selection strategies and retention ratios for both models.

Method	Retention	GQA Accuracy (%)	
		LLaVA	Qwen2.5
No compression	100%	65.9	36.1
Random dropping	70%	65.1	35.7
Uniform pooling	70%	65.3	34.7
Attention ($l=2$)	70%	64.2	34.4
Attention ($l=15$)	70%	66.4	35.3
Random dropping	50%	63.9	31.7
Uniform pooling	50%	63.9	33.1
Attention ($l=2$)	50%	61.8	32.9
Attention ($l=15$)	50%	65.3	34.5
Random dropping	25%	60.7	29.0
Uniform pooling	25%	61.7	30.4
Attention ($l=2$)	25%	56.2	29.9
Attention ($l=15$)	25%	64.4	31.6
Attention inverse ($l=15$)	25%	50.2	26.7

achieves 64.4%—only 1.5 pp below the 65.9% full baseline at roughly one-fifth the compute—while random dropping falls to 60.7% and uniform pooling to 61.7% under the same budget. At $r=0.50$, $l=15$ reaches 65.3%, matching the full baseline to within 0.6 pp while halving computational cost. Notably, attention at $l=2$ (56.2% at $r=0.25$) performs worse than both query-agnostic baselines, underscoring that the source layer is a critical determinant of pruning quality.

Qwen2.5-VL-3B.

On Qwen2.5-VL-3B the absolute accuracy values are lower (baseline 36.1%) owing to the smaller model capacity, but the cross-method rank ordering is consistent with LLaVA. Attention at $l=15$ again leads, reaching 31.6% at $r=0.25$ and 35.3% at $r=0.70$, 0.8 pp below the full baseline. As on LLaVA, $l=2$ underperforms random dropping at higher retention ratios (34.4% vs. 35.7% at $r=0.70$), confirming that early-layer attention provides a weaker token selection signal regardless of architecture.

Comparison with query-agnostic baselines.

The advantage of attention-based pruning over query-agnostic strategies is concentrated at aggressive retention rates. At $r=0.70$, random dropping, uniform pooling, and attention at $l=15$ produce near-identical results on both models, as sufficient redundant context remains to absorb any selection differences. At $r=0.25$, the gap widens substantially: on LLaVA, $l=15$ outperforms random dropping by 3.7 pp and uniform pooling by 2.7 pp, and on Qwen by 2.6 pp and 1.2 pp respectively. This pattern confirms that query-conditioned relevance scores from Equation (1) provide a more effective selection criterion than content-agnostic strategies precisely when the selection decision is most consequential.

Effect of retention ratio.

Figure 5 plots accuracy against retention ratio for all strategies across both models. At high retention ($r \geq 0.7$), all methods remain within a narrow band of the no-compression

baseline, as sufficient visual context is preserved regardless of selection strategy. Methods begin to diverge below $r = 0.5$, with the gap most pronounced at $r = 0.25$, where the selection criterion becomes the primary determinant of whether task-relevant evidence is retained. We therefore treat $r = 0.25$ as the most diagnostic operating point throughout our analysis. At $r = 0.1$, all methods become unstable and no strategy reliably preserves accuracy, suggesting a practical lower bound on safe token removal for general visual question answering.

Efficiency-accuracy trade-off.

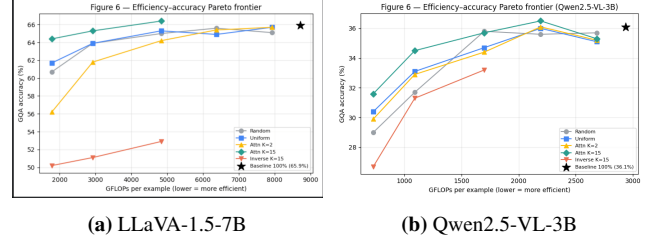


Figure 1: Efficiency-accuracy Pareto frontiers on GQA. Middle-layer attention ($l=15$) defines the upper bound across both architectures. Inverse-attention pruning consistently occupies the lower bound, confirming that top-attended tokens are genuinely necessary rather than incidentally salient.

Figure 1 plots GQA accuracy against theoretical FLOPs for all strategies across both models. Attention at $l=15$ defines the upper bound of the efficiency-accuracy frontier at every operating point: on LLaVA it achieves 64.4% accuracy at approximately one-fifth the FLOPs of the full baseline, a level unreached by any other compressed strategy. The inverse probe occupies the lower bound on both models, with a 14.2 pp gap below $l=15$ on LLaVA and a 4.9 pp gap on Qwen at $r=0.25$. The larger gap on LLaVA suggests the attention signal is a more selective criterion on the larger, more expressive model.

6. Analysis

6.1. Effect of Aggressive Pruning

The effect of token selection becomes substantially more visible as the visual token retention rate decreases (in Appendix A.2). At high retention ratios ($r \geq 0.7$), most evaluated methods preserve sufficient visual evidence for a wide array of queries, leading to relatively small performance deltas between distinct pruning strategies. Under aggressive pruning regimes, however, the specific choice of selection criterion serves as a critical intervention, directly determining whether task-relevant semantic evidence remains available to the multi-modal language backbone.

The $r = 0.25$ retention setting is particularly informative because it exposes stark differences in relevance estimation across strategies without completely collapsing downstream model performance. On GQA, the performance split between pruning strategies becomes highly pronounced at this threshold. For LLaVA, middle-layer attention pruning ($l=15$) re-

tains an impressive 65.3% accuracy—retaining the bulk of the 65.9% uncompressed baseline performance while discarding three-quarters of the visual sequence. In contrast, early-layer attention selection ($l=2$) drops to 61.8%, while the causal probe of inverse pruning ($l=15$) falls sharply to 50.2%. A remarkably similar trend replicates on Qwen2.5-VL-3B, where layer-15 attention achieves 34.5% accuracy compared to 32.9% for layer-2 attention and a baseline floor of 26.7% under inverse pruning conditions.

These results yield two vital mechanistic insights. First, middle-layer attention maps contain significantly more useful, query-aware relevance information than early visual blocks. Second, the dramatic performance gap supports the claim that the attention ranking signal carries meaningful relevance information: tokens assigned highly salient scores are genuinely necessary for downstream VQA reasoning rather than being incidentally prioritized.

Remarkably, our data supports an even stronger architectural claim: the structural layout and depth of the selection signal matters more than the raw quantity of preserved tokens. For instance, evaluating LLaVA under severe compression at $r = 0.10$ using middle-layer selection ($l=15$) yields 64.4% accuracy, handily outperforming an allocation with $2.5\times$ more visual bandwidth ($r = 0.25$) guided by an early-layer ($l=2$) score (61.8%).

However, at the absolute limits of compression ($r = 0.10$), performance trajectories across all methods become visibly unstable. Even our optimal middle-layer attention pruning begins to lose substantial performance. This degradation suggests an empirical lower bound to visual compression; hyper-aggressive pruning eventually removes relational and spatial evidence required for reliable reasoning, even when the remaining token subsets are highly prioritized by semantic attention masks. Overall, these dynamics demonstrate that pruning efficacy relies heavily on balancing target token importance with basic scene coverage constraints.

6.2. Question-Type Breakdown

Figure 2 breaks down GQA accuracy at $r=0.25$ into four semantic categories for LLaVA-1.5-7B. Counting is excluded as the evaluated subset did not yield sufficient examples for a reliable per-method estimate. The breakdown reveals that pruning reliability is not uniform across question types—attention-based selection is highly effective for localised evidence but degrades on queries requiring distributed spatial context. Results for Qwen2.5-VL-3B follow the same rank ordering and are provided in Appendix A.3.

Object recognition.

Attention at $l=15$ reaches 73.0%, within 2 pp of the 75.0% full-token baseline, while random dropping falls 5 pp short at 70.0%. The inverse probe drops sharply to 51.5%, a 21.5 pp gap below $l=15$, confirming that the top-attended tokens for object questions are spatially concentrated and causally necessary.

Attribute identification.

All methods cluster between 65.0–66.0%, closely matching

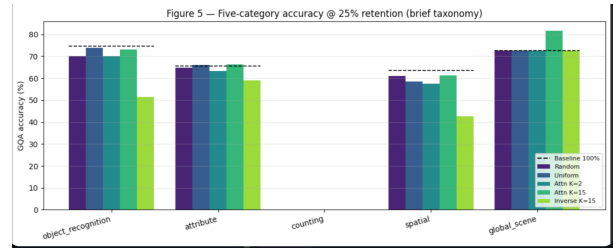


Figure 2: Per-category GQA accuracy at $r=0.25$ for LLaVA-1.5-7B. Middle-layer attention ($l=15$) outperforms all baselines on object and attribute categories, matches them on global scene, and remains the strongest compressed method on spatial reasoning despite an overall degradation. Counting is excluded due to insufficient sample count. Qwen2.5-VL-3B results are in Appendix A.3.

the 66.0% baseline. This tight grouping indicates that at-tribute questions can be resolved from a small number of locally distinctive tokens regardless of selection strategy, making the attention signal less critical for this category.

Spatial reasoning.

Spatial questions are the most fragile category under aggressive pruning. Even $l=15$ (61.0%) falls 3 pp below the 64.0% baseline, and the inverse probe collapses to 43.0%, yielding the largest causal gap observed across all categories. Notably, early-layer attention ($l=2$, 58.0%) underperforms random dropping (61.0%), confirming that a weak attention signal is more harmful than no signal at all when the task requires preserving multiple distributed regions and their relative arrangement.

Global scene understanding.

Attention at $l=15$ reaches 81.0%, 8 pp above the 73.0% full-token baseline, while random dropping, uniform pooling, and the inverse probe all track the baseline closely. We interpret this as evidence that global scene categories can be resolved from a compact subset of primary regions, and that the remaining tokens introduce noise that middle-layer attention pruning effectively suppresses. This finding must be interpreted cautiously: the global-scene category contains only 11 examples in our evaluated subset, and we do not generalise beyond the observed trend.

6.3. Qualitative examples

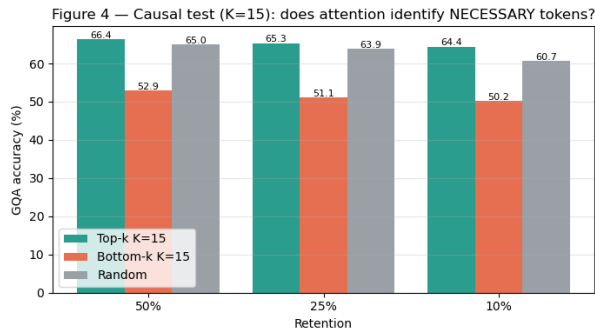
The prediction files also reveal characteristic success and failure patterns. For localized object queries, attention retention often preserves the decisive region. For example, questions asking whether a table or couch is present, or which furniture is behind a dog, are answered correctly under top-attention retention but become unreliable under inverse pruning. This is consistent with the view that high-attention visual tokens preserve object-level evidence.

Failures often involve relations, comparisons, or subtle attributes. A question such as whether two objects have the same color requires attending to both objects and comparing their attributes. If pruning preserves only one salient object, the answer may flip even though the retained tokens look reasonable in isolation. Similarly, spatial questions such as whether one object is to the right of another require layout

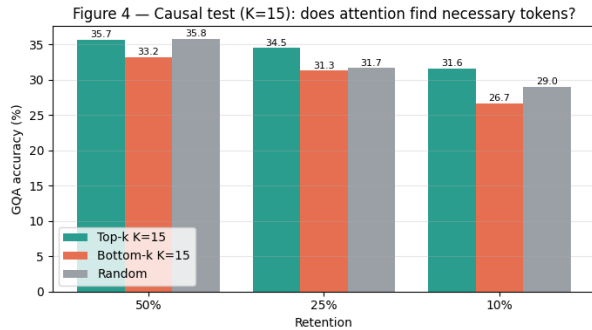
context. These examples explain why category-level averages differ: the usefulness of attention depends on whether the answer can be supported by a compact subset of visual tokens.

Table 4 summarizes the resulting interpretation. The table is not a substitute for evaluation; rather, it expresses the mechanism suggested by the results. Token pruning is most faithful when visual evidence is compact and least faithful when evidence is distributed or relational.

6.4. Inverse pruning



(a) LLaVA-1.5-7B Causal Probe



(b) Qwen2.5-VL-3B Causal Probe

Figure 3: Causal test ($K=15$): does attention identify necessary tokens? Top- K retention is compared against Bottom- K (inverse) and Random retention at 50%, 25%, and 10% retention. The persistent gap between Top- K and Bottom- K at low retention rates demonstrates that high-attention tokens carry causally meaningful evidence.

Inverse pruning is not meant to be a practical baseline; it is a causal sanity check on the scoring rule. The severe drop under inverse layer-15 pruning shows that tokens assigned low attention by the middle layer are not interchangeable with high-attention tokens. This strengthens the claim that attention contains information about useful visual evidence. However, it does not prove that attention is a complete causal explanation. It only shows that the ranking is informative under this intervention.

7. Ablation Studies

7.1. Layer Depth on Visual Relevance Estimation

The selection of the specific Transformer layer utilized to extract the routing token attention weights serves as our

primary structural ablation. Figure 4 displays the downstream GQA task accuracy across representative index layers ($l \in \{2, 8, 15, 22\}$) evaluated under a rigid 25% token retention threshold.

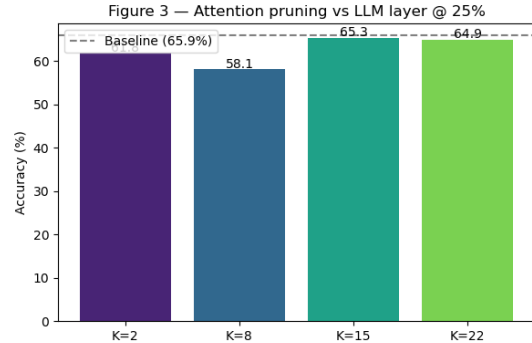


Figure 4: LLaVA-1.5-7B task accuracy (%) on GQA across distinct source attention layers (l) at a 25% token retention constraint. The dashed horizontal line benchmarks the uncompressed baseline performance (65.9%).

Our empirical results reveal that attention-derived token importance is not a static, low-level property of the input image; rather, informative visual routing scores emerge dynamically as the visual sequence matches and cross-attends with the text prompt through deep structural layers. At a 25% retention operating point on LLaVA, utilizing attention weights from a mature intermediate layer ($l = 15$) preserves 65.3% accuracy, remaining within a minimal 0.6 percentage point (pp) margin of the uncompressed baseline. Conversely, relying on early-stage representations ($l = 2$) drops performance significantly to 58.1%, while an extraction at $l = 8$ marks the performance floor at 61.8%.

Mechanistically, these findings suggest a clear division of labor across the multi-modal text-image processing pipeline. Early layers ($l \leq 8$) primarily register localized visual saliency, high-frequency edges, or rigid positional bias masks, failing to effectively bind the specific semantic intent of the text query to relevant image coordinates. Intermediate layers ($l = 15$), however, operate at a higher level of abstraction, incorporating cross-modal textual context to better reflect query-conditioned visual relevance.

Interestingly, as sequence processing reaches late saturation layers ($l = 22$), selection quality begins to trace a slight downward trajectory (64.9%), showing that attention weights in the final segments of the model become overly optimized for token generation dynamics rather than localized visual preservation. This layered non-linearity confirms that raw attention scores from an arbitrary transformer block cannot be uniformly treated as a reliable proxy for token explanation or pruning suitability.

8. Discussion

8.1. Does attention identify semantically useful tokens?

The results support a qualified answer. Attention-derived relevance is useful, but not universally reliable. Across

LLaVA-1.5-7B and Qwen2.5-VL-3B, middle-layer attention provides a stronger pruning signal than early-layer attention, suggesting that visual-token relevance is layer-dependent. The inverse-pruning results further show that the attention ranking is behaviourally meaningful under the pruning intervention: retaining low-attention tokens performs substantially worse than retaining high-attention tokens at the same retention rate. However, this does not prove that attention is a complete causal explanation of model reasoning.

The competitiveness of random and uniform baselines in some regimes shows that part of the robustness to pruning comes from redundancy in the visual token sequence rather than perfect query-conditioned grounding. Therefore, attention should be interpreted as a noisy but useful relevance signal, not as a faithful explanation by default. The main scientific insight is conditionality: attention-based pruning works best when the query depends on compact or localized visual evidence, and weakens when useful evidence is distributed, relational, spatial, or underrepresented in the retained tokens.

8.2. Implications for efficient MLLMs

The practical implication is that efficient MLLMs should not rely on a single global pruning rule. Object and attribute questions may tolerate aggressive token reduction, while spatial, counting, OCR, or comparison questions may require higher retention or diversity constraints. This suggests that future pruning systems should adapt the retention rate or scoring rule based on query type, visual complexity, or model uncertainty.

The scientific implication is that pruning experiments can serve as behavioural probes of internal representations. A retained high-attention token is not automatically an explanation, but comparing top-attention, inverse-attention, random, and uniform retention connects token rankings to downstream behaviour. This provides a stronger diagnostic test than attention visualization alone.

8.3. Relation to prior pruning claims

Prior work often emphasizes large reductions in token count or FLOPs with limited average accuracy loss. Our results refine this interpretation. Average robustness to pruning can arise from genuine query-conditioned relevance, redundancy in visual tokens, dataset biases, or answer priors. The competitiveness of random and uniform selection indicates redundancy, while the failure of inverse pruning indicates that attention still carries useful signal. Both findings are needed for a balanced interpretation.

8.4. Why middle layers may work better

A plausible explanation is that early layers are dominated by local visual patterns, token position, and low-level feature interactions. Middle layers appear to encode stronger alignment between the language query and relevant visual regions, making them more suitable for ranking visual tokens under aggressive pruning. Late layers may become more shaped by answer-generation dynamics, although this requires further

testing. Thus, layer selection is not merely a hyperparameter; it provides insight into where useful multimodal grounding may emerge.

8.5. Limitations

Several limitations remain. First, the experiments use controlled subsets rather than full benchmark evaluations, so the results should not be interpreted as definitive leaderboard scores. Second, GFLOPs are analytical compute proxies and do not capture hardware-level latency. This is especially important for LLaVA, where pruning is implemented through attention masking rather than physical token removal.

Third, the analysis focuses mainly on GQA-style examples with a smaller MMBench sanity check. Broader claims require larger and more diverse benchmarks, especially counting, OCR, spatial reasoning, and multi-object comparison tasks. Fourth, the attention score aggregates across heads and text positions, which may hide head specialization. Fifth, attention-based pruning tests whether attention is useful under intervention, but it does not establish full causal semantic necessity. Gradient-based attribution, attention rollout, counterfactual masking, or object-level ablations would strengthen this claim.

A stronger version of the study would also include paired bootstrap intervals or McNemar tests over matched examples. Finally, architecture generality remains open because LLaVA-style self-attention, Qwen2.5-VL dynamic visual processing, and explicit cross-attention architectures may define attention-derived relevance differently.

9. Conclusion

This paper studied query-aware visual token pruning as a diagnostic probe of multimodal attention. Across LLaVA-1.5-7B and Qwen2.5-VL-3B, the results support a conditional version of the hypothesis: attention-derived, query-conditioned relevance scores can identify useful visual tokens better than query-agnostic baselines, but only under certain conditions. Middle-layer attention produced the strongest proxy efficiency-accuracy frontier and substantially outperformed inverse-attention pruning, suggesting that attention rankings contain behaviourally useful relevance information.

However, early-layer attention was unreliable, random and uniform baselines remained competitive at higher retention rates, and spatial reasoning was more fragile under aggressive pruning. These findings show that attention-based pruning is neither merely an efficiency trick nor a complete explanation of model reasoning. Instead, it is a useful but imperfect probe of how MLLMs allocate visual evidence for a query. Future work should extend this analysis to larger MLLMs, explicit cross-attention architectures, counting and OCR-heavy tasks, and causal region-level interventions.

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801 A. Extended Results

802 This appendix provides complete result tables, retention
 803 curve figures, and per-category breakdowns that support
 804 the main paper. Section A.1 gives full accuracy tables for all
 805 retention ratios on both GQA and MMBench. Section A.2
 806 shows the full retention curve trajectories. Section A.3 pro-
 807 vides the Qwen2.5-VL-3B category breakdown referenced
 808 in Section 6.2. Section A.4 summarises the mechanistic
 809 interpretation of per-category behaviour.

810 A.1. Full Accuracy Tables

811 Tables 2 and 3 extend Table 1 by reporting all five retention
 812 ratios $r \in \{0.90, 0.70, 0.50, 0.25, 0.10\}$ for both models.
 813 The GQA table includes inverse pruning at $r=0.25$ as the
 814 causal probe result. The MMBench table reports Qwen2.5-
 815 VL-3B only, as MMBench serves as a cross-dataset sanity
 816 check rather than a primary evaluation benchmark; LLaVA
 817 MMBench results were not collected for the same reason.
 818 Bold entries indicate the best compressed method at each
 819 retention block.

Table 2: Full GQA accuracy (%) across all retention ratios for both models. Bold indicates the best compressed result within each retention block.

Method	Retention	GQA Accuracy (%)	
		LLaVA	Qwen2.5
No compression	100%	65.9	36.1
Random dropping	90%	65.1	35.7
Uniform pooling	90%	65.7	35.1
Attention ($l=2$)	90%	65.7	35.2
Attention ($l=15$)	90%	—	35.3
Random dropping	70%	65.6	35.6
Uniform pooling	70%	64.9	36.0
Attention ($l=2$)	70%	65.4	36.1
Attention ($l=15$)	70%	—	36.5
Random dropping	50%	65.0	35.8
Uniform pooling	50%	65.3	34.7
Attention ($l=2$)	50%	64.2	34.4
Attention ($l=15$)	50%	66.4	35.7
Random dropping	25%	63.9	31.7
Uniform pooling	25%	63.9	33.1
Attention ($l=2$)	25%	61.8	32.9
Attention ($l=15$)	25%	65.3	34.5
Attention inverse ($l=15$)	25%	50.2	26.7
Random dropping	10%	60.7	29.0
Uniform pooling	10%	61.7	30.4
Attention ($l=2$)	10%	56.2	29.9
Attention ($l=15$)	10%	64.4	31.6

Table 3: Full MMBench accuracy (%) across all retention ratios for Qwen2.5-VL-3B. MMBench serves as a cross-dataset sanity check; LLaVA MMBench results were not collected as the primary evaluation focus is GQA. Bold indicates the best compressed result within each retention block.

Method	Retention	MMBench Acc. (%)
		Qwen2.5-VL-3B
No compression	100%	81.5
Random dropping	90%	81.5
Uniform pooling	90%	80.0
Attention ($l=2$)	90%	80.0
Attention ($l=15$)	90%	79.5
Random dropping	70%	80.5
Uniform pooling	70%	79.5
Attention ($l=2$)	70%	79.0
Attention ($l=15$)	70%	80.0
Random dropping	50%	76.5
Uniform pooling	50%	78.5
Attention ($l=2$)	50%	79.0
Attention ($l=15$)	50%	80.0
Random dropping	25%	76.0
Uniform pooling	25%	75.5
Attention ($l=2$)	25%	76.0
Attention ($l=15$)	25%	76.5
Random dropping	10%	74.0
Uniform pooling	10%	71.5
Attention ($l=2$)	10%	74.0
Attention ($l=15$)	10%	72.5

A.2. Retention Curve Trajectories

Figure 5 plots GQA accuracy against the token retention ratio for all strategies across both models. The curves confirm the two key observations stated in Section 6: methods converge at $r \geq 0.70$ and diverge sharply below $r = 0.50$, with $r=0.25$ as the most diagnostic operating point.

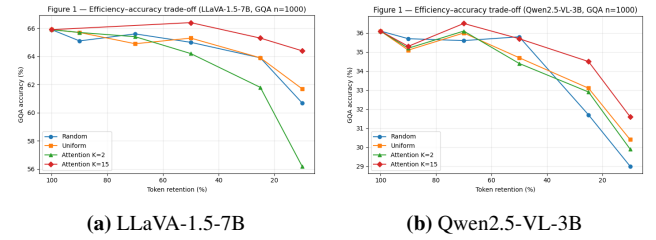


Figure 5: GQA accuracy (%) as a function of retention ratio r for all strategies. Methods remain within a narrow band at high retention and diverge below $r=0.50$. At $r=0.10$, all methods become unstable.

A.3. Question-Type Breakdown: Qwen2.5-VL-3B

Figure 6 shows per-category GQA accuracy at $r=0.25$ for Qwen2.5-VL-3B. The rank ordering across methods is consistent with LLaVA-1.5-7B in all four categories, confirming that the qualitative findings in Section 6.2 are not architecture-specific. Absolute accuracy values are lower

832 throughout owing to the smaller model capacity.

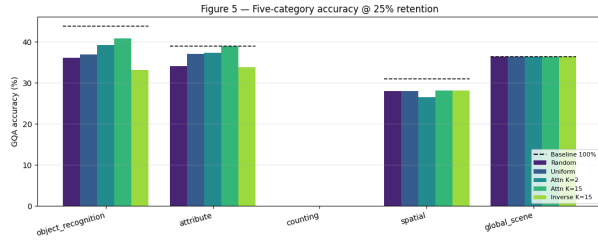


Figure 6: Per-category GQA accuracy at $r=0.25$ for Qwen2.5-VL-3B. Rank ordering is consistent with LLaVA-1.5-7B (Figure 2).

833 **Object recognition.** Attention at $l=15$ achieves 41.0%
834 against a 44.0% baseline. Both random dropping (36.0%)
835 and the inverse probe (33.1%) fall further behind, preserving
836 the same rank ordering as LLaVA.

837 **Attribute identification.** Layer-15 attention (39.0%)
838 matches the 39.0% baseline exactly, while random dropping
839 (34.0%) and the inverse probe (34.0%) both fall 5 pp
840 short—a larger gap than on LLaVA, indicating that the atten-
841 tion signal is more discriminative on the smaller model for
842 this category.

843 **Spatial reasoning.** All compressed methods converge
844 between 26.0–28.2% against a 31.0% baseline. The conver-
845 gence suggests that at $r=0.25$, the spatial relational context
846 required for these questions is disrupted regardless of selec-
847 tion strategy.

848 **Global scene understanding.** All methods including
849 the inverse probe converge between 36.0–36.5% against a
850 37.0% baseline. In contrast to LLaVA, middle-layer attention
851 provides no additional benefit, suggesting that Qwen extracts
852 global scene information from a broader token distribution
853 rather than a compact salient subset.

854 A.4. Mechanistic Interpretation by Question Type

855 Table 4 summarises the expected pruning behaviour per ques-
856 tion type, derived from the empirical findings in Sections 6.2
857 and A.3. It is intended as a concise reference rather than a
858 substitute for the quantitative results above.

Table 4: Expected pruning behaviour by question type under aggressive token compression ($r=0.25$), with mechanistic justification.

Question type	Expected behaviour	Reason
Object	Strong	Evidence is spatially concentrated; top- K reliably covers the target region.
Attribute	Moderate / strong	Local cues suffice on large models; attention is critical on smaller models.
Spatial	Weak / moderate	Requires multiple distributed regions and compact token subsets lose relational context.
Counting	Uncertain / weak	Evidence is distributed and fine-grained; small sample count prevents a reliable estimate.
Global scene	Mixed	Large models benefit from selective attention; smaller models saturate on any representative subset.